MBU DIGITAL LIBRARY · PRACTICE LEARNING RESOURCE

SQL

SQL101 | Suggested sequence · Semester 2 | Student-authored study guide

How to use these notes

Portable SQL essentials for querying, joining, grouping, modifying, and reasoning about relational data. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · SELECT, filtering, and sorting

SELECT names columns, FROM names a relation, WHERE filters rows, and ORDER BY sorts the result. SQL uses three-valued logic: NULL is unknown, so test it with IS NULL rather than = NULL.

Worked example

```
SELECT product, price FROM items WHERE price >= 100 ORDER BY price DESC;
```

Check your understanding

- Why does column = NULL not match NULL rows?
- Does ORDER BY happen before or after WHERE?

MODULE 02 · Joins and aggregation

Joins combine rows using related columns. COUNT/SUM/AVG aggregate groups; GROUP BY defines groups and HAVING filters groups after aggregation. Choose join type according to whether unmatched rows should remain.

Worked example

```
SELECT department_id, COUNT(*) FROM students GROUP BY department_id HAVING COUNT(*) > 10;
```

Check your understanding

- When should HAVING be used instead of WHERE?
- What rows does LEFT JOIN preserve?

MODULE 03 · Data changes and safe queries

INSERT adds rows, UPDATE changes matching rows, DELETE removes them. Preview a predicate with SELECT and use a transaction for related changes. Parameterized queries keep values separate from SQL syntax.

Worked example

```
UPDATE inventory SET quantity = quantity - 1 WHERE item_id = $1 AND quantity > 0;
```

Check your understanding

- Why is a parameterized query safer?

• What is the risk of UPDATE without WHERE?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.